How the three ensembles come to agree

An exact study of a 2D quantum lattice gas

Prasang Das Aryan Shakya Dhruv Thakur

The question

A gas can be described three ways:

- **microcanonical** — fix energy and particle number
- **canonical** — fix temperature, exchange energy with a bath
- **grand-canonical** — also exchange particles

They agree *in the thermodynamic limit*. Any finite system — a trap, a simulation — is not there yet.

What we measure

How the agreement sets in as the lattice grows, and how interactions change it.

Prasang

Model and method

Prasang

The model: hard-core Bose–Hubbard

$$\hat{H} = -t \sum_{\langle i,j \rangle} \left(\hat{b}_i^\dagger \hat{b}_j + \text{h.c.} \right) + V \sum_{\langle i,j \rangle} \hat{n}_i \hat{n}_j$$

- bosons hop between neighbors (t)
- repel or attract neighbors (V)
- at most one boson per site

- cold atoms in an optical lattice
- energy in units of t; $k_B = 1$

Not the quantum lattice-gas *automaton* — same name, different thing.

Prasang

Exact diagonalization

- $\hat{H}$ conserves N: block diagonal, one block per sector
- diagonalize each block *fully* — the partition sums need every level
- sizes $L = 2, 3, 4$ ($M = 4, 9, 16$ sites); half-filled block is the cost ceiling

One spectrum $\{E_{N,\alpha}\}$, built once and cached, feeds all three ensembles.

Prasang

Ensembles and what to measure

Aryan

Three ensembles, one spectrum

Microcanonical

$$\Omega(E, N; \Delta E)$$

$$S = \ln \Omega$$

Canonical

$$Z_N = \sum_{\alpha} e^{-\beta E_{N,\alpha}}$$

$$C_V = \beta^2 \mathrm{Var}(E)$$

Grand

$$\Xi = \sum_{N} e^{\beta \mu N} Z_N$$

$$\partial_\mu \langle N \rangle = \beta \, \mathrm{Var}(N)$$

Each is a different weighting of the same levels.

Aryan

Equivalence is a fluctuation

Canonical weight is a Laplace transform of the count:

$$Z_N = \int dE\, e^{S(E)-\beta E}.$$

The saddle at $S'(E^*) = \beta$ ties the two temperatures together; its width gives

$$\frac{\sigma_E}{|\langle E\rangle|} \sim \frac{1}{\sqrt{M}}, \qquad \frac{\sigma_N}{\langle N\rangle} \sim \frac{1}{\sqrt{M}}.$$

These ratios are the small parameters of equivalence — and smooth spectral sums, so they survive on a tiny spectrum.

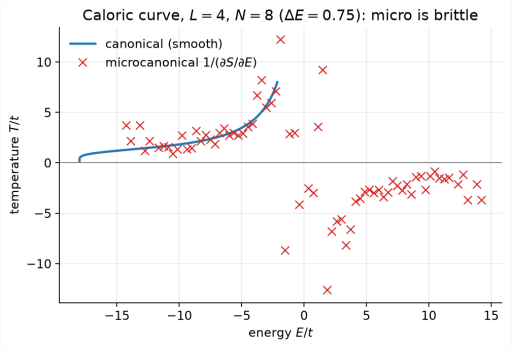
Aryan

Why not read T off the entropy?

$1/T = \partial S/\partial E$ is the honest definition — and useless here.

- $S = \ln \Omega$ is a jagged histogram
- slope diverges in gaps
- flips sign past the band center (negative T)

So we track the fluctuations instead.



Aryan

Does it compute the right thing?

Checked against independent references:

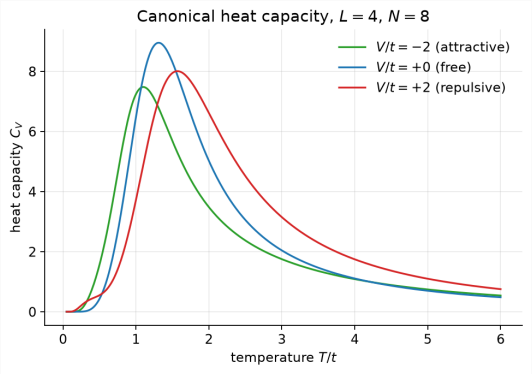
- one-particle band matches $-2t[\cos k_x + \cos k_y]$ to 10^{-15}
- ground states match a sparse Lanczos solver
- $C_V = \beta^2 \mathrm{Var}(E)$, $\partial_\mu \langle N \rangle = \beta \mathrm{Var}(N)$ vs numerical derivatives (10^{-5})
- $T \to 0, \infty$ limits of S and C_V

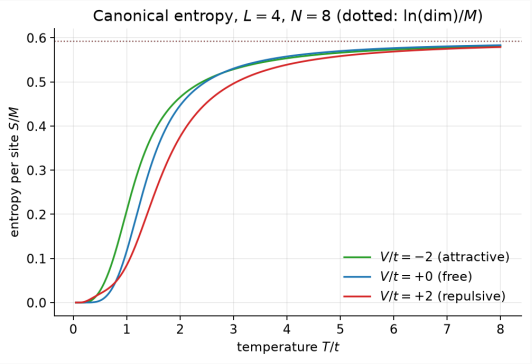
Aryan

Results

Dhruv

Thermodynamics behaves

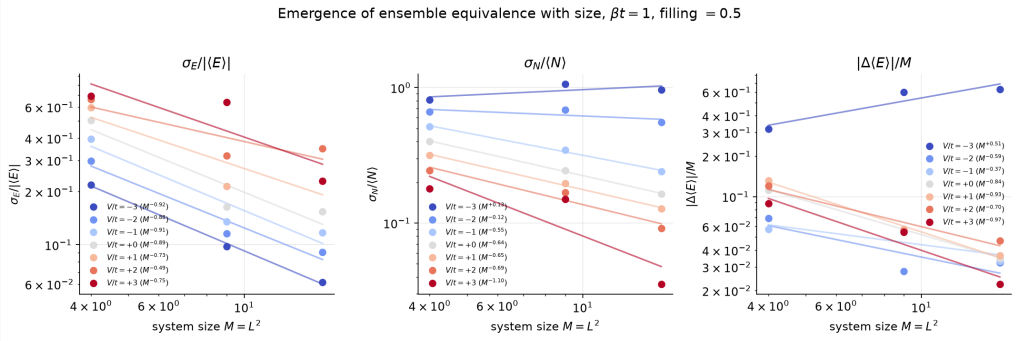




Schottky-like heat capacity; entropy climbs from 0 to $\ln(\dim)/M$. Interaction shifts the scale, not the shape.

Dhruv

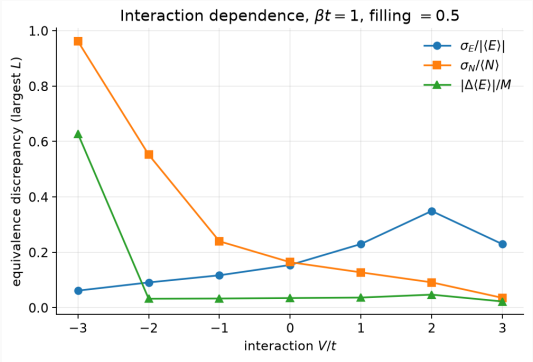
Equivalence emerges with size



Free and repulsive: all three discrepancies fall with M, near $M^{-1/2}$.

Dhruv

Attraction breaks it — in the number channel



V/t	$p(\sigma_N/\langle N\rangle)$	$\sigma_N/\langle N\rangle$
-3	$+0.13$	0.96
0	-0.64	0.16
$+3$	-1.10	0.04

At $V/t = -3$ the exponent is **positive**: the mismatch *grows* with size — the gas wants to phase separate.

Dhruv

Takeaway

The ensembles converge for free and repulsive particles, and diverge for attractive ones — in the particle number, where phase separation lives.

Measured through fluctuations, because the microcanonical thermometer fails on small bounded spectra.

Thank you

Prasang Das **Aryan Shakya** **Dhruv Thakur**

Hard-core Bose–Hubbard model, exact diagonalization